

Question #1 of 19

Question ID: 1494703

A company is *most likely* to have a high cost of capital if the firm has a low:

A) total debt-to-EBITDA ratio.



B) interest coverage (IC) ratio.



C) debt-to-equity (D/E) ratio.



Explanation

Low interest coverage (IC) suggests that a firm has a reduced ability to service additional debt. Holding business risk constant, companies with higher proportions of debt in their capital structure are likely to face higher costs of capital.

(Module 17.2, LOS 17.f)

Question #2 of 19

Question ID: 1480222

When attempting to build a risk premium into the required returns of stocks in a developing country, an analyst should use the:

A) country's weighted average cost of capital.



B) country spread model.



C) modified Gordon Growth model.



Explanation

The country spread model uses data from a developed market, then adjusts it using the difference between the bond yields for the emerging and developed markets. Neither a modified Gordon Growth model nor a weighted average cost of capital will do this job.

(Module 17.2, LOS 17.d)

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Question ID: 1480215

The equity risk premium is the difference between:

A) estimated equity returns and estimated bond returns.



B) the estimated equity return and the risk-free return.



C) the required equity return and the risk-free return.



Explanation

The equity risk premium reflects the return in excess of the risk-free rate that investors require for holding stocks. It is derived by subtracting the risk-free return from the required return.

(Module 17.1, LOS 17.a)

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Question ID: 1480225

Junior analyst Quentin Haggard is struggling with a required return calculation. His main concern is compensating for exchange rate fluctuations between the country where his company is based and the home country of a portfolio of stocks he is analyzing. Haggard should calculate the return in his home country's currency, then adjust:

A) for expected changes in the foreign country's currency value.



B) for expected changes in the foreign country's inflation rate.



C) the beta to account for exchange-rate fluctuations.



Explanation

The proper method of compensating for changes in exchange rates is to calculate the required return in the home currency, then adjust the return using forecasts for changes in the exchange rate.

(Module 17.2, LOS 17.d)

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Question ID: 1480218

Currently the market index stands at 1,190.45. Firms in the index are expected to pay cumulative dividends of 35.71 over the coming year. The consensus 5-year earnings growth forecast for these firms is expected to increase to 6.2% up from last year's forecast of 4.5%. The long-term government bond is yielding 5.0%. According to the Gordon growth model, what is the equity risk premium?

A) 4.2%.



B) 1.2%.



C) 2.5%.



Explanation

Equity risk premium = $(35.71 / 1,190.45) + (6.2\%) - 5.0\% = 4.2\%$

(Module 17.2, LOS 17.c)

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Question ID: 1480223

Candace Elwince is attempting to calculate the required return of Skeun Inc., a machine-tool manufacturer in a small Eastern European country. Elwince has solid data from the German market but is not sure how to account for the exchange-rate risk Skeun investors would face. Her *best* choice for creating a risk premium is the:

A) Gordon Growth model.



B) difference between the inflation rates of both markets.



C) difference between the bond yields of both markets.



Explanation

The country spread model suggests an analyst can approximate the risk premium between a developed market and an emerging market by subtracting the bond yields in the developed market from yields in the emerging market.

(Module 17.2, LOS 17.d)

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Question ID: 1494697

Estimates of historical ERP resulting from backfilling of index data is *most likely* to:

A) bias the estimate upward.



B) result in an unbiased estimate.



C) bias the estimate downward.



Explanation

Backfilling (going back in historical time to obtain data after index constituents are identified) would most likely result in survivorship bias—and, therefore, bias the estimate of ERP upward.

(Module 17.2, LOS 17.c)

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Question ID: 1480217

Ben Jacobs, CFA, is attempting to calculate a historical equity risk premium. His first estimate uses geometric mean equity returns and long-term bond yields. His second estimate uses arithmetic mean returns and short-term bond yields. The effect of the changes in methodology in the second estimate, relative to the first, will:

- A) both increase the size of the risk premium.
- B) both decrease the size of the risk premium.
- C) have offsetting effects.

**Explanation**

Switching from a geometric mean to an arithmetic mean will increase the mean equity return. All else being equal, that will increase the estimated risk premium. When the yield curve slopes upward, short-term bonds yield less than long-term bonds. Thus, the equity risk premium estimate will be larger when short-term bond rates are used.

(Module 17.2, LOS 17.c)

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Question ID: 1480219

Types of estimates of the equity risk premium are *least likely* to include:

- A) extemporized estimates.
- B) macroeconomic model estimates.
- C) ex-ante estimates.

**Explanation**

There are four types of estimates of the equity risk premium: historical estimates, forward-looking (ex-ante) estimates, macroeconomic model estimates, and survey estimates.

(Module 17.2, LOS 17.c)

Sofie Johansen is a new financial analyst at Rikdom Investments. Johansen meets with her manager to discuss a possible investment in Helsevesen Health Care. Johansen's manager asks her to estimate the cost of common equity using the Fama-French five-factor model for Helsevesen as a first step in valuing Helsevesen.

Johansen estimates Helsevesen's cost of equity by regressing Helsevesen's excess return on relevant risk factors using past 120-month returns. The resulting factor betas and risk premiums are shown in **Factor Betas and Risk Premiums**.

Factor Betas and Risk Premiums

Factor	Factor Beta	Risk Premium
Market (ERP)	1.40	3.5%
Size (SMB)	0.85	1.8%
Value (HML)	0.65	1.5%
Profitability (RMW)	0.15	1.6%
Investment (CMA)	0.60	2.1%

Johansen decides to use the government 20-year benchmark interest rate of 3.4% as the risk-free rate.

Johansen's manager also asks her to estimate Helsevesen's cost of equity using the bond yield plus risk premium (BYPRP) method. For this estimate, Johansen assumes a historical risk premium of 5.4% earned by equity investors over long-term corporate bond yields. Further, she calculates an estimated cost of debt of 4.9% using matrix valuation.

Finally, Johansen is asked to estimate the ERP for the country of Yukon. Selected data is given in **Yukon Selected Data**

Yukon Selected Data

	Per Year
Estimated earnings growth rate for public companies	5.00%
Forecasted market P/E growth	1.20%
Real GDP growth rate forecast	2.50%
Estimated aggregate dividend yield	0.60%
Current level of inflation	3.50%
ST government yield	5.00%
LT government yield	5.50%
Growth in shares outstanding	1.35%

Yukon's central bank, with a tight monetary policy, is expected to bring inflation down to 2.50% per year.

Question #10 - 13 of 19

Question ID: 1586154

Helsevesen's cost of common equity using the CAPM is *closest* to:

- A) 12.3%.
- B) 10.3%.
- C) 8.3%.



Explanation

Helsevesen's estimated cost of common equity using the CAPM can be calculated as follows:

$$r_e = r_f + \beta(\text{ERP})$$

$$r_e = 0.034 + 1.40(0.035) = 0.0830, \text{ or } 8.3\%.$$

(Module 17.1, LOS 17.e)

Question #11 - 13 of 19

Question ID: 1586155

Helsevesen's cost of common equity using the Fama-French five-factor model is *closest* to:

- A) 10.3%.
- B) 12.3%.
- C) 8.3%.



Explanation

Helsevesen's estimated cost of common equity using the Fama-French five-factor model can be calculated as follows:

$$r_e = r_f + \beta_1(\text{ERP}) + \beta_2\text{SMB} + \beta_3\text{HML} + \beta_4\text{RMW} + \beta_5\text{CMA}$$

$$r_e = 0.034 + 1.40(0.035) + 0.85(0.018) + 0.65(0.015) + 0.15(0.016) + 0.60(0.021) = 0.1231, \text{ or } 12.3\%.$$

(Module 17.1, LOS 17.e)

Question #12 - 13 of 19

Question ID: 1586156

Helsevesen's cost of common equity using the bond yield plus risk premium (BYPRP) model is *closest* to:

A) 8.3%.



B) 10.3%.



C) 12.3%.



Explanation

Helsevesen's estimated cost of common equity using the BYPRP model can be calculated by adding the estimated cost of debt of 5.8% Johansen derived from matrix pricing to Johansen's estimated premium of 6.2% earned by equity investors relative to long-term corporate bond yields:

$$r_e = r_d + RP$$

$$r_e = 0.049 + 0.054 = 0.103, \text{ or } 10.3\%.$$

(Module 17.1, LOS 17.e)

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Question ID: 1494702

Using the Grinold-Kroner model, the estimated ERP for Yukon is *closest* to:

A) 2.65%.



B) 3.15%.



C) 7.65%.



Explanation

$i = 2.50\%$ (long-term estimate after central bank's action)

$G = 2.50\%$ (real GDP growth rate)

$\Delta P/E = 1.20\%$

$DY = 0.60\%$

$\Delta S = 1.35\%$

$R_f = 5.50\%$ (LT government rate)

$ERP = [DY + i + G + \Delta P/E + \Delta S] - R_f = [0.60 + 2.50 + 2.50 + 1.20 + 1.35] - 5.50 = 2.65\%.$

(Module 17.2, LOS 17.c)

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Question ID: 1494696

If a company's debt is publicly traded, the *most appropriate* estimate of its cost of debt can be derived from:

- A) matrix pricing, based on the yields on traded securities with the same maturity and credit ratings. 
- B) an inferred credit rating on the debt based on the financial ratios of the company. 
- C) the yield to maturity for the firm's longest-maturity straight debt outstanding. 

Explanation

If a company's debt is publicly traded, the yield to maturity for the firm's longest-maturity straight debt outstanding is our best estimate of its cost of debt. If a company's debt is not traded (or is thinly traded), we can use matrix pricing, based on the yields on traded securities with the same maturity and credit ratings. If the debt is not credit rated, we can use financial ratios of the company such as interest coverage or financial leverage to infer a credit rating on the debt.

(Module 17.1, LOS 17.b)

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Question ID: 1480221

In the process of estimating beta for a private company, unlevering the beta calculated for the publicly traded comparable company accomplishes what goal?

- A) Establishing a baseline level of leverage. 
- B) Improving the accuracy of the estimate in the event that the private company's debt is of low quality. 
- C) Isolating market risk. 

Explanation

Market risk, also known as systematic risk, is the risk common to all assets within a certain class. Deleveraging the beta strips out the company-specific risk related to the target company's leverage, thereby isolating market risk. Beta calculations do not require a baseline level of leverage. The equation for calculating beta for private companies assumes the company in question has high-grade debt. The deleveraging process will not help if the assumption is incorrect.

(Module 17.2, LOS 17.d)

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Question ID: 1480220

Equity analyst Yasmine Cordova of Substantial Securities is trying to determine the investment appeal of shares of Maxwell Mincemeat, a small food company. Cordova has assembled the following data about the company:

- Internal rate of return: 9.4%.
- Maxwell's 20-year bond yield to maturity: 7.9%.
- Maxwell's two-year bond yield to maturity: 6.1%.
- Treasury bill yield: 3.4%.
- Maxwell's estimated beta: 2.1.
- Maxwell's 20-year bonds are priced at \$102.65.
- Maxwell's two-year bonds are priced at \$101.47.
- Estimated return of Russell 2000 Index: 12.3%.
- Substantial's credit analyst estimates that Maxwell's equity warrants a premium of 4.9% over its bonds.

Cordova wants to make sure her estimates are accurate, so she decides to calculate the estimated required return in two ways. She opts for the bond-yield plus risk premium method and the capital asset pricing model. To check her work, she wants to compare the estimates derived under each method. The difference between the required returns is *closest* to:

A) 9.29%.



B) 5.89%.



C) 5.30%.



Explanation

The capital asset pricing model uses the following equation:

$$\text{Required return} = \text{risk-free rate} + \text{beta} \times \text{equity risk premium}$$

To calculate the required return under CAPM, use the Russell 2000 index return, the beta, and the risk-free rate.

$$\text{Required return} = 3.4\% + 2.1 \times (12.3\% - 3.4\%) = 22.09\%.$$

The bond-yield model uses the following equation:

$$\text{Required return} = \text{yield to maturity on long-term bonds} + \text{risk premium.}$$

$$\text{Required return} = 7.9\% + 4.9\% = 12.8\%.$$

The difference between the two estimated required returns is 9.29%.

(Module 17.2, LOS 17.d)

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Question ID: 1494695

Bottom-up factors that affect a company's cost of capital are *most likely* to include:

- A) asset nature and liquidity. 
- B) capital availability and market conditions. 
- C) legal and regulatory considerations and tax jurisdiction. 

Explanation

Top-down (i.e., macro) factors that affect the cost of capital include capital availability, market conditions, legal and regulatory considerations, and tax jurisdiction.

Bottom-up (i.e., company-specific) factors that affect the cost of capital include business or operating risk, asset nature and liquidity, financial strength and profitability, and security features.

(Module 17.1, LOS 17.a)

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Question ID: 1480224

The country risk rating model:

- A) determines a risk premium for any foreign market. 
- B) determines a risk premium for an emerging market. 
- C) depends on forecasts of exchange rates. 

Explanation

The country risk rating model begins with a model from a developed country, then modifies that model with inputs from an emerging market to derive a risk premium for the emerging market. Forecasts of exchange rates may well be part of the model, but they are not a requirement.

(Module 17.2, LOS 17.d)

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Question ID: 1480216

An analyst attempting to derive the equity risk premium for a stock starting from the required return for that stock would find which of the following statistics *least* useful?

- A) The stock's beta. 
- B) The stock's estimated return. 

C) Historical 10-year Treasury bond rates.



Explanation

The required return for a stock is equal to the risk-free return plus beta times the equity risk premium. An analyst starting from the required return would need beta and a risk-free rate. Historical 10-year T-bond rates can be used as an estimate of the risk-free rate. Since the analyst is starting with the required return, estimated returns are not needed.

(Module 17.2, LOS 17.c)